ASSIGNMENT – 4 (CRITICAL)

CSC-303 – Computer Networks

The Quantum Internet: Enhancing Classical Internet Services One Qubit at A Time

Group 4 -

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1. Overview of the Work

The research paper “The Quantum Internet: Enhancing Classical Internet Services One Qubit at A Time” by Angela Sara Cacciapuoti et al.,addresses the complex and interdependent relationship between the classical Internet and the emerging Quantum Internet.

A common misconception is that the Quantum Internet will eventually replace the classical Internet. However, the authors argue that the Quantum Internet heavily relies on the classical Internet for essential services, such as signaling and coordination.

The paper proposes a paradigm shift: viewing the interplay not as unidirectional but bidirectional. The Quantum Internet has the profound potential to support and fundamentally enhance classical Internet functionalities across multiple protocol layers.

2. Key Outcomes

The main goal of this research is to show how integrating quantum resources into the classical protocol stack can overcome existing fundamental limitations.

- **Physical Layer (Bit Quantum Transmission):**

  The paper shows that shifting to quantum carriers allows networks to overcome classical information bottlenecks. By exploiting phenomena like superadditivity and the quantum switch, classical bits can be transmitted over noisy channels, achieving data rates that surpass classical capacity limits.

- **Data Link Layer (Channel Quantum Access Control):**

  Contention resolution on shared channels is historically plagued by collisions. The research demonstrates that by utilizing a specific class of multipartite entangled states (W states), nodes can randomly and fairly elect a leader without any classical signaling, providing a distributed, collision-free access method.

- **Network Layer (Quantum Multipath Routing):**

  The authors introduce the concept of coherent multipath routing. having a well-defined phase relationship between different possibilities of a quantum state directly defines **quantum coherence**. This coherence allows a single classical packet encoded in a quantum carrier to be transmitted simultaneously across a quantum superposition of multiple paths, entirely bypassing the classical need to duplicate the packet (which is forbidden by the quantum no-cloning theorem).

- **Upper Layers (Quantum Security Service):**

  To secure communications against future quantum computer attacks, the paper highlights the use of Quantum Key Distribution (QKD) and Quantum Secure Direct Communication (QSDC) to generate encrypting keys, radically enhancing the confidentiality of classical transport and application layers.

3. Novelty and Novel Points

The paper is unique in how it rethinks the architecture of the Quantum Internet, shifting it from a top-heavy application to an integrated cross-layer enhancement tool.

- **The Bidirectional Interplay Model:**

  Both the Classical Internet and Quantum Internet provide services to each other.e.g. in **quantum teleportation**, the Quantum Internet depends on the Classical Internet for classical communication and signaling, showing **Classical** → **Quantum** interaction. On the other hand, in **Entanglement Access Control**, quantum entanglement is used to improve access to a shared classical communication channel, showing **Quantum** → **Classical** interaction.

  Showing interplay is bidirectional.

- **Entanglement as an Access Control Mechanism:**

  Uses Entanglement to decide which node gets access to a shared communication channel. W states allow the nodes to select one node as the leader, so only that node transmits. This avoids collisions because multiple nodes do not transmit simultaneously, and it also handles the **hidden node problem**. The paper describes this as a strategy that does not require classical signaling protocols like CSMA or further coordination after the measurement.

- **Distributed Quantum Computing for Routing:**

  The proposal to use quantum algorithms to solve complex classical routing optimization problems (which scale exponentially for classical hardware) represents a novel convergence of quantum computing and classical network layer operations.

4. Technologies and Protocols

To implement the ideas presented in this paper, specific quantum technologies, protocols, and resources are required:

- **Quantum Teleportation:**

  A strategy used to transmit a qubit without physically moving the particle that stores it.

  It requires two types of communication resources: a quantum resource (a pair of maximally entangled qubits shared between the source and destination) and a classical resource (a pair of bits sent from the source to the destination).

  The process is carried out using a sequence of quantum functionalities—specifically entanglement generation and distribution, followed by local quantum operations at both the source and destination—interleaved with the transmission of a classical message.

- **Superdense Coding:**

  A quantum communication protocol that allows two bits of data to be transmitted by "coding" them into a single qubit, assuming that the sender and receiver already share a pre-shared entangled resource.

  In this networking model, packets received from the data link layer can either be encoded and sent traditionally through the classical physical layer, or routed to a quantum super-physical layer to be encoded using the superdense protocol.

  From the perspective of the classical Internet, superdense coding acts as a complex feature of the physical layer that provides a traditional bit transmission service.

- **Entanglement Access Control Protocol :**

  A protocol adapted from advanced quantum management to solve the channel contention problem (deciding which node gets to speak next) on a classical communication channel// principles of

  How it works:

  Each of n nodes performs a single local measurement on its respective W-state qubit.

  Exactly one random node observes outcome '1' (equal probability for all), while the rest observe '0'.

  The node with '1' becomes the elected leader with exclusive transmission rights, while nodes with '0' remain silent—instantly solving collisions, avoiding the hidden-node problem, and keeping the leader's identity private.

- **W state:**

  An entanglement state exhibiting the ability of fairly and randomly electing a leader among a set of nodes.

- **Quantum Key Distribution (QKD):**

  A security protocol that exploits the fundamental laws of quantum mechanics to allow communicating nodes to generate shared, random cryptographic keys.

  How it works:

  Parties exchange photons or quantum states over a channel.

  According to quantum mechanics, any unauthorized eavesdropper attempting to intercept and measure these particles will inevitably disturb their states, introducing detectable errors.

  The legitimate parties check for errors to ensure the channel is secure, discard corrupted data, and use the error-free remainder to create an identical, information-theoretically secure key for encrypting classical messages.

5. Gaps Identified in This Work

While the theoretical foundation is robust, the authors identify several severe technical challenges that currently hinder practical deployment.

- **The Hardware Limitations (The NISQ Gap):**

  We are currently in the Noisy Intermediate-Scale Quantum (NISQ) era. The proposed enhancements require fault-tolerant quantum processors, which do not yet exist.

- **The Synchronization Gap:**

  Reliable entanglement generation and distribution to remote nodes require extremely tight synchronization and signaling. The current classical Internet operates on a "best-effort" nature, which is unlikely to satisfy these strict timing requirements.

- **The Architectural Interface Gap:**

  Classical networking relies on the "separation of concerns" (e.g., the OSI model). However, quantum enhancements require cross-layer interactions (e.g., data link access control requires quantum physical connectivity). There is currently no standardized global interface to manage these complex dependencies.

6. Overall Conclusion

The authors conclude that the design of the Quantum Internet cannot be conducted in a vacuum or modeled simply as an application layer on top of classical infrastructure. Because the Quantum Internet offers profound enhancements to classical network services—from physical data rates to application security—designing its protocol stack requires a deep understanding of cross-layer bidirectional dependencies. Overcoming the current hurdles will require a multidisciplinary collaborative effort spanning quantum physics, communications engineering, and network architecture.

7. Anticipated Solution (Unified Classical-Quantum Interface)

To bridge the architectural gap identified in the paper, the anticipated solution is the development of a **Unified Classical-Quantum Interface**.

- **The Concept:**

  Instead of strictly isolating layers according to the OSI model, this unified interface would act as a cross-layer management plane.

- **How it Works:**

  It would dynamically allocate quantum resources (like entanglement) to whichever classical layer requires it most urgently—whether routing algorithms at the network layer or MAC contention at the data link layer.

- **Addressing Synchronization:**

  By building this interface directly into hardware controllers, networks could bypass the "best-effort" latency of standard classical routing, establishing the dedicated, tightly synchronized classical signaling channels necessary for reliable entanglement swapping and distribution.